

2.1.

$$\left[\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right]_B = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} \text{ t.j. } \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 1 & 3 & x_1 \\ 2 & 7 & x_2 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 3 & x_1 \\ 0 & 1 & x_2 - 2x_1 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{cc|c} 1 & 0 & 7x_1 - x_2 \\ 0 & 1 & x_2 - 2x_1 \end{array} \right)$$

slejnó my $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$

$$\left[\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right]_B = \begin{pmatrix} 7x_1 - x_2 \\ -2x_1 + x_2 \end{pmatrix}; \left[\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right]_B = \begin{pmatrix} 7y_1 - y_2 \\ -2y_1 + y_2 \end{pmatrix}$$

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle \stackrel{\text{Ponkornilov}}{=} \left[\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right]_B \cdot \left[\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right]_B =$$

$$= (7x_1 - x_2)(7y_1 - y_2) + (-2x_1 + x_2)(-2y_1 + y_2)$$